GA-CNN: Convolutional Neural Network Based on Geometric Algebra for Hyperspectral Image Classification

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***Abstract*— Convolutional neural networks (CNNs) have achieved state-of-the-art performance in hyperspectral images (HSIs) classification, which is widely used for the analysis of remotely sensed images. HSI includes spectral and spatial information from several hundreds of spectral data channels. Recent CNN models deal with various bands of HSIs as independent channels, which may lead to the loss of dependencies between different channels or the loss of associated information between each channel and the global. This article proposes a novel CNN model based on geometric algebra (GA), dubbed GA-CNN, to process the HSIs in a holistic way without losing the interrelationship among channels. Specifically, taking advantage of GA, different band images are represented as GA multivectors to capture the inherent structures and preserve the correlation of those channels. In particular, all the basic modules of our model, such as convolutional layers and the backpropagation algorithm, are extended to the GA domain. We evaluate the performance of the proposed GA-CNN model in classification tasks on four well-known HSI datasets. The experimental results indicate that our GA-CNN model outperforms traditional and state-of-the-art real-valued CNNs with higher classification accuracy and fewer model parameters.**

***Index Terms*— Convolutional neural network (CNN), geometric algebra (GA), hyperspectral images (HSIs) classification, multivectors.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain abundant spectral bands with spatial features and image information. Due to the high resolution of spectral dimension, HSIs can provide a lot of information about different land-cover objects, which makes HSI classification become a hot spot in remote sensing community, widely used for agricultural monitoring [1], ecological management [2], and other civil and military fields [3]. Nowadays, the convolutional neural network (CNN) becomes a state-of-the-art technique for HSI classification in deep learning (DL)-based classification methods [4].

At present, the research on DL for HSI classification focuses mainly on training samples and network model design. As for training samples, DL normally needs sufficient training samples to ensure a satisfactory performance, and it is difficult to obtain a large number of manually labeled samples for HSI classification. Establishing a DL model for HSI classification with few labeled samples is a current hot topic of research. DL, transfer learning, and lightweight models have been proposed to deal with the problem of small-sample sets [5]. Li et al. [6] proposed a spectral context-aware transformer segmentation (SCAT) method, in which the transformer model was introduced to better capture spectral–spatial information and to effectively improve the segmentation outcome of HSI. Regarding the network model design, spectral feature-based classification and spectral–spatial-feature-based classification are the two categories in CNNs. Zhang et al. [7] proposed the unsupervised spectral–spatial CNN model, which considers the spectral–spatial features and better captures the feature information of the sample; 1-D CNN was first proposed in [8] using spectra features of all bands, which consists of one convolutional layer and two fully connected layers to achieve better accuracy compared with traditional SVM using the radial basis function as the kernel (RBF-SVM) [9]. In order to improve the accuracy of classification, 2-D CNN [10], [11] pays more attention to extracting spatial feature information, while 3-D CNN [12], [13], [14], [15] focuses on extracting spectral–spatial feature information. Because using just 2-D CNN may result in missing channel relationship information and using just 3-D CNN may make the model very complex, Guo et al. [16] and Roy et al. [17] combined 2-D CNN and 3-D CNN. Zhong et al. [18] first applied residual blocks to the HSI classification, and the residual blocks use identity mapping to connect each other 3-D convolutional layer. Jia et al. [19] used the multiscale characteristics of 3-D Gabor kernels to characterize spectral–spatial information and adopted a multibranch network structure to reduce the number of network layers. Then, by considering consecutive spectral and spatial squeeze-and-excitation residual blocks, a lightweight spectral–spatial squeeze-and-excitation residual

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network (S3EResBoF) [20] was proposed to better characterize the HSI channelwise spectral–spatial features. Considering that not all bands provide the same contribution on a CNN and spatial information from all pixels in the neighborhood not contributes equally to a CNN, attention mechanism was introduced into deep convolutional networks. An end-to-end residual spectral–spatial attention network (RSSAN) [21] was designed by a spectral attention module and a spatial attention module, which enhanced the contribution of surrounding pixels with the same labels as the center pixel on classifying the center pixel while weakening the contribution of those with different labels. Ge et al. [22] proposed an HSI classification method based on an adaptive hash attention mechanism and a lower triangular (AHA-LT) network, which achieved spectral attention module and the adaptive hash spatial attention module in series. However, the attention mechanism is artificially determined and cannot represent the structural characteristics of HSI. These models generate good classification maps when using enough training samples, but the contradiction between the increased model parameters and the limited available training samples has not been solved. At the same time, nearly, all real-valued CNNs for HSI classification just treat each channel independently, which leads to the loss of structural characteristics among different bands in HSIs.

Actually, in order to preserve the correlation between multidimensional signals, extensive studies on neural networks based on different algebraic systems have been undertaken. Aizenberg [23] and Hirose [24] proposed complex-valued neural networks with multivalued neurons, which show the remarkable properties of multivalued neurons that are used as basic units in neural networks. The backpropagation (BP) algorithm for complex-valued neural networks was proposed by Leung and Haykin [25], Benvenuto and Piazza [26], Nitta [27], and Georgiou and Koutsougeras [28], which provides a new paradigm for updating the complex-valued neural network and has been successfully applied in the classification of complex signals. Hänsch and Hellwich [29] first employed complex-valued CNNs for object detection in polarimetric synthetic aperture radar images. Although their proposed network model consists of only one convolutional layer, the experimental results are superior to those of complex-valued multilayer perceptrons. In order to describe more multidimensional signals, complex-valued neural networks have been explored in higher dimensional space. Nitta [30], [31] extended neural networks to the quaternion domain and further proposed a version of the BP algorithm based on quaternion. Kominami [32] introduced multivalued CNNs with multi-input and multioutput and verified the improved learning ability on a handwritten digital image dataset. Meanwhile, in image processing [33], [34], [35], quaternion has become a widespread tool for color images by encoding color channels in a holistic manner. However, quaternion only has four parts and thus cannot handle hundreds of channels in HSIs.

Consider that geometric algebra (GA) allows the processing of geometric entities such as points and lines. It is a very efficient language for solving many tasks connected to the design of intelligent systems [36], [37], [38], [39]. In particular, spatiotemporal interest points (STIPs) based on GA have achieved satisfactory performance in human action recognition [38], which represents multichannel videos as GA multivectors and provides a new strategy to build multiscale space based on GA. Speeded-up robust feature (SURF) based on GA is a novel algorithm to process multispectral images [39]. Through taking advantages of GA, the efficiency of extracting feature in multispectral images is better than other feature extraction algorithms. Therefore, applying geometric neural computing theory to HSI images and establishing a powerful model-based method are the main goals of this thesis.

In particular, each pixel in an HSI is represented as a multivector, and accordingly, the image is represented as a GA matrix rather than several independent real-valued matrices. Taking the GA matrix as the input of our network, we design a series of basic modules, e.g., GA convolutional layer and GA fully connected layer. While the traditional real-valued convolution is only capable to enforce scaling transformation on the input, specifically, the GA convolution achieves the scaling and the rotation of input in the color space, which provides us with more structural representation. Based on these modules, we can establish GA-CNN to process HSIs in a more effective way.

The major contributions of this article are given as follows.

1) We propose a series of novel CNN models based on GA (1D-GACNN that only processes spectral features and 3D-GACNN that processes spectral–spatial features, collectively called GA-CNN), which extend conventional CNN from the real number domain to the GA domain.
2) The proposed GA-CNN is capable of preserving more structural information among multiple channels by representing HSIs as GA multivectors.
3) The BP algorithm based on GA (GA-BP) algorithm is deduced in detail to train the proposed GA-CNN model.
4) Classification experiments on four well-known datasets show that the proposed GA-CNN offers a superior performance compared to real-valued CNNs with respect to higher classification accuracy and fewer model parameters.

The rest of this article is organized as follows. Section II introduces the basics of GA and neural networks based on GA. Section III describes the proposed GA-CNN in detail. Comparison experimental results between GA-CNN and real-valued CNNs are provided in Section IV, followed by concluding remarks drawn in Section V.

II. Preliminary

A. Basics of Geometric Algebra

GA [40], [41] was first described by William K. Clifford, also called Clifford algebra. For multidimensional signals, GA is not only an effective framework to handle the representation and computation issues but also a useful tool for widespread use in mathematics and physics [42], [43], [44], [45], [46], [47], [48], [49].

Mathematically, suppose that $\mathbb{G}_n$ denotes a 2^n-dimensional vector space, and there exist a set of orthogonal bases $\{e_1, e_2, \ldots, e_n\}$. The power set $\boldsymbol{\gamma}$ of $\{1, \ldots, n\}$ can turn the

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basis into an ordered one with the index set $\mathbf{b}$

$$\{\mathbf{b} = (a_1, \ldots, a_r) \in \boldsymbol{\gamma}, 1 \le a_1, \ldots, a_r \le n\}. \tag{1}$$

Then, the basis of $\mathbb{G}_n$ is denoted by

$$\left\{e_A := e_{a_1}, \ldots, e_{a_r} | A \in \mathbf{b}\right\}. \tag{2}$$

For example, the basis in the 2^3 vector space can be described as

$$\left\{e_\varnothing, e_1, e_2, e_3, e_{12}, e_{13}, e_{23}, e_{123}\right\}. \tag{3}$$

For convenience, in the rest of this article, $a_1, \ldots, a_r$ will be denoted by $a_{1,\ldots,r}$, and $e_\varnothing = 1$.

In general, the multiplication in GA will follow the following rules:

$$\begin{cases} e_{a_i}^2 = 1, & a_i = 1, \ldots, p \\ e_{a_i}^2 = -1, & a_i = p+1, \ldots, p+q \\ e_{a_i} e_{a_j} = -e_{a_j} e_{a_i}, & a_i \ne a_j \end{cases} \tag{4}$$

and $\mathbb{G}_n$ can also be denoted as $\mathbb{G}_{p,q}$, with $2^n = p + q$.

Set $\mathbb{G}_n^t$ represents the t-vector part of $\mathbb{G}_n$ as follows:

$$\mathbb{G}_n^t = \{e_A | e_A \in \mathbf{b}, |A| = t\}. \tag{5}$$

An arbitrary element of the GA is given by

$$x = \sum_{t=0}^{n} \langle x \rangle_t = \sum_A [x]_A e_A \tag{6}$$

where $[x]_A \in \mathbb{R}$ represents the value of each component of the multivector. For example, an element in the 2^3 vector space can be represented as

$$\begin{aligned} x &= \langle x \rangle_0 + \langle x \rangle_1 + \langle x \rangle_2 + \langle x \rangle_3 \\ &= x_0 + x_1 e_1 + x_2 e_2 + x_3 e_3 + x_{12} e_{12} + x_{13} e_{13} \\ &\quad + x_{23} e_{23} + x_{123} e_{123}. \end{aligned} \tag{7}$$

The addition in GA can be defined as

$$x + y = \sum_A ([x]_A + [y]_A) e_A. \tag{8}$$

The geometric product in GA can be written in the following form:

$$x \otimes_{p,q} y = x \cdot y + x \wedge y \tag{9}$$

where $x \cdot y$ and $x \wedge y$ represent the inner and outer products in GA, respectively.

For performing more complex multivector operations, several rules are defined as follows.

1) The inversion of GA is

$$x^\dagger = \langle x \rangle_t^\dagger = (-1)^{\frac{t(t-1)}{2}} \langle x \rangle_t. \tag{10}$$

2) The conjugation of GA is

$$x^* = \langle x \rangle_t^* = (-1)^{\frac{t(t+1)}{2}} \langle x \rangle_t. \tag{11}$$

3) The module of GA is

$$|\langle x \rangle| = \sqrt{\sum_{k=0}^{n} |\langle x \rangle_t|^2} \tag{12}$$

$$|\langle x \rangle_t| = \sqrt{\langle x \rangle_t \cdot \langle x \rangle_t}. \tag{13}$$

B. Neural Networks Based on Geometric Algebra

A GA neuron (GAN) [50] can be viewed as a mapping function from $(\mathbb{G}_{p,q})^n$ to $\mathbb{G}_{p,q}$

$$y = \sum_{i=1}^{n} w_i \otimes_{p,q} x_i + \theta \tag{14}$$

where $x_i \in \mathbb{G}_i$, $w_i \in \mathbb{G}_i$, and $\theta \in \mathbb{G}_i$ denote the inputs, weights, and bias, respectively.

Then, a basic GAN (BGAN) is derived from a GAN in the case of $n = 1$, and a mapping function from $\mathbb{G}_{p,q}$ to $\mathbb{G}_{p,q}$ is given by

$$y = w_i \cdot x_i + \theta. \tag{15}$$

The activation function is an essential component of a neural network. A real-valued activation function [50] is used for all multivector outputs in the GA-CNN model. More specifically, for any multivector $M \in \mathbb{G}_n$, the real-valued activation function is defined

$$f(M) = \sum_{A \in \mathbf{b}} [M]_A e_A. \tag{16}$$

III. CNNs Based on Geometric Algebra

For each real kernel, it only sums up the outputs corresponding to different channels and ignores the complicated interrelationship between them. Therefore, on one hand, the important structural information of color is lost and we may not obtain the optimal representation of the HSI image. On the other hand, too many degrees of freedom to the learning of real convolution kernels result from simply summing up the outputs, and thus, they may be at a high risk of overfitting even if imposing heavy regularization terms.

In this section, we extend CNN from the real-time domain to the GA domain. The overall structure of our network is shown in Fig. 1. In our proposed GA-CNN, inputs, outputs, and weights are represented by GA multivectors. Each multichannel pixel in the HSI image is first integrated into a multivector and then processed with GA convolution kernels. After the GA pooling and the fully connected layers, the output labels correspond to the input pixels. The operations in each layer and the training algorithm will be introduced in detail in the following.

A. Input Preparation

Each pixel $I(x, y)$ of an HSI can be represented as a multivector in $\mathbb{G}_n$

$$\begin{aligned} I(x,y) = \sum_{0 \leqslant i \leqslant n} [I(x,y)]_i e_i + \sum_{1 \leqslant i < j \leqslant n} [I(x,y)]_{ij} e_{ij} \\ + \cdots + [I(x,y)]_{1,\ldots,n} e_{1,\ldots,n} \end{aligned} \tag{17}$$

where $[I(x, y)]_i$, $[I(x, y)]_{ij}$, and $[I(x, y)]_{1,\ldots,n}$ are the spectral components of the HSI pixel point I at the position (x, y). Essentially, the HSI pixel is mapped in GA space using a GA multivector.

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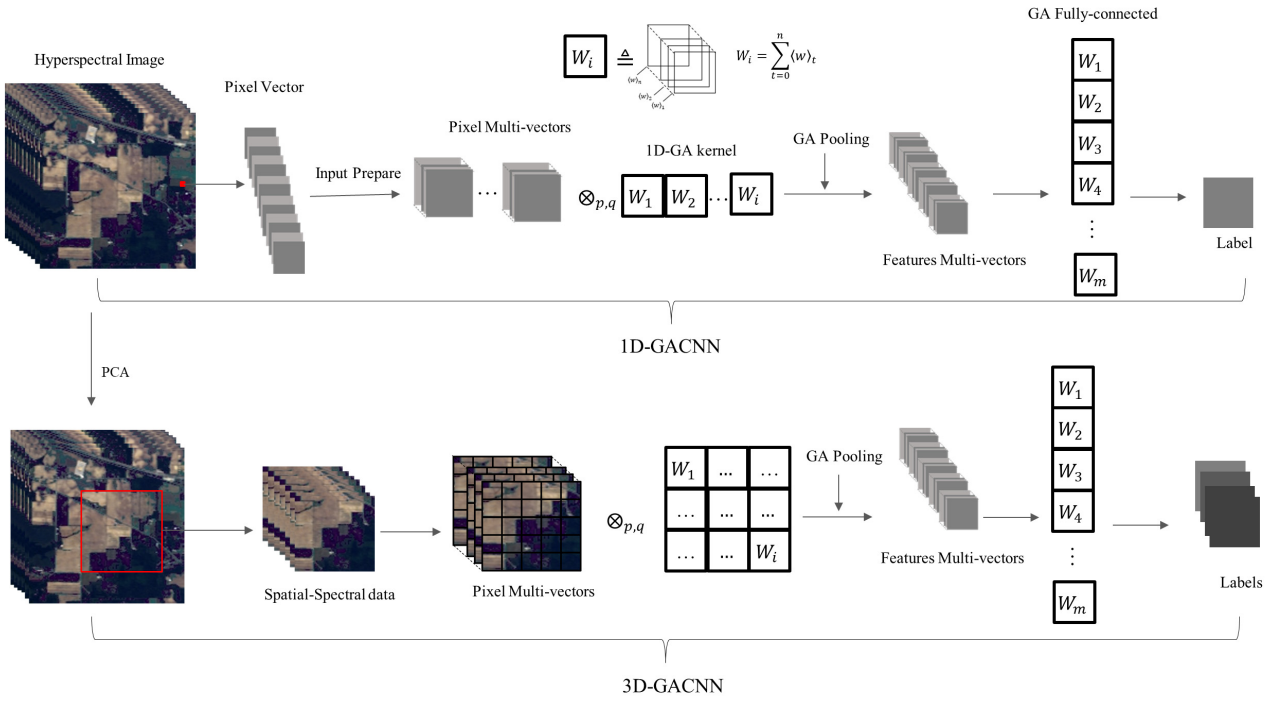


Fig. 1. Schematics of the proposed GA-CNN.

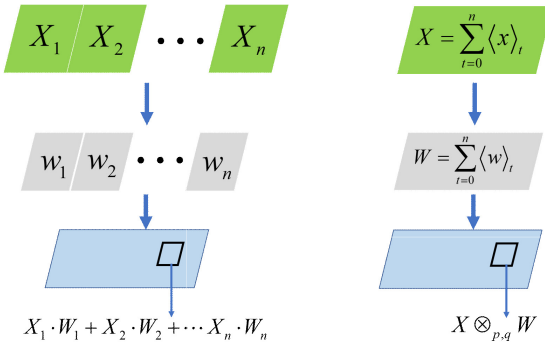


Fig. 2. (a) Real-valued and (b) GA convolutions.

B. Convolutional Layers

Convolutional layers with multiple kernels can extract multichannel for multidimensional signals. Compared with real-value CNNs, a convolutional layer based on GA has multiple kernels, as shown in Fig. 2. The output of the jth neuron in the lth convolutional layer is

$$y_j^{l_{\text{conv}}} = \sum_{A\in\mathbf{b}} g^{(l)}\left(\sum_i \left(\mathrm{w}_{kj}^{(l)} \cdot x_k^{(l-1)} + \theta_j^{(l)}\right) e_A\right) \tag{18}$$

where $x_k^{(l-1)}$ represents the kth input from the previous layer and $\mathrm{w}_{kj}^{(l)}$ represents the convolution kernels.

C. GA Pooling Layers

In the GA-CNN, the GA-based pooling layer is an extension of the real-valued pooling as follows:

$$\begin{aligned}&\operatorname{softmax}_\alpha\left(\{x_k\}_{k=1}^n\right)\\ &= \sum_{A\in\mathbf{b}} \frac{\sum_k x_{kA} e^{\alpha x A}}{\sum_j e^{\alpha x_{kA}} e_A}\\ &= \begin{cases} \sum_{A\in\mathbf{b}} \left(\max_k x_{kA}\right) e_A, & \alpha \to \infty \\ \sum_{A\in\mathbf{b}} \left(\frac{1}{n}\sum_k x_{kA}\right) e_A, & \alpha \to 0, x_{kA} \in \mathbb{R} \\ \sum_{A\in\mathbf{b}} \left(\min_k x_{kA}\right) e_A, & \alpha \to -\infty \end{cases}\end{aligned} \tag{19}$$

where $\forall \alpha \in \mathbb{R}, \{x_k\}_{k=1}^n \in \mathbb{G}_n$. In short, it performs the corresponding real-valued pooling operation on each component of the multivector.

D. GA Fully Connected Layers

As mentioned above, the previous layers of GA-CNN can extract more interrelationship features than those of real-valued CNNs. In order to keep this important information, we propose the multilayer perceptron based on GA (GA-MLP). Its operation is the same as that of the previous layer to maintain the internal correlation information between channels. Specifically, similar to the real CNN, its fully connected layer can be considered a special case of 1-D convolutional layer. Its core has the same shape as the input, and our proposed GA fully connected layer follows the same rules. GA-MLP can be viewed as the output of fully connected layers that are connected to all neurons in the last pooling layer. The GA-MLP is defined as follows:

$$\begin{aligned} s_j^{(L)} &= \sum_k w_{kj} \cdot x_k^{(L-1)} + \theta_j^{(L)} \\ y_j^{(L)} &= \sum_{A\in\mathbf{b}} g\left(\left[s_j^{(L)}\right]_A\right) e_A \\ &= \sum_{A\in\mathbf{b}} g\left(\sum_k \left[w_{kj} \cdot x_k^{(L-1)} + \theta_j^{(L)}\right]_A\right) e_A \end{aligned} \tag{20}$$

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where L represents the output layer and k represents the kth input in the $(L-1)$th layer.

E. BP Algorithm

The BP algorithm provides an effective means to compute the steepest gradient descent on an error function $E(\cdot)$ with chain rule for training the network. In this article, a GA-BP is deduced and applied to train the proposed GA-CNN.

The standard output square error is

$$E = \frac{1}{2}\sum_{j}\sum_{A\in\mathbf{b}}\left(\left[d_j\right]_A - \left[y_j^{(L)}\right]_A\right)^2 \tag{21}$$

where $y_j \in \mathbb{G}_n$ represents the output and $d_j \in \mathbb{G}_n$ represents the desired output in the output layers.

The derivation of weights updating can be found in APPENDIX.

IV. Experiments and Discussion

In this section, the performance of the proposed GA-CNN (1D-GACNN and 3D-GACNN) is evaluated on four different datasets against those of traditional and state-of-the-art real-valued CNNs. The difference between the GA-CNN and real-valued CNN lies in the change of data expression and calculation method. All the experiments are performed by using Python 3.6 on the Ubuntu 16.04.3 LTS operating system with the GeForce GTX 1080 GPU and 10.92-GB RAM. In addition, as the most widely used quality metrics [16], the overall accuracy (OA), the average accuracy (AA), and the Kappa coefficient (K) are utilized to evaluate the performances of the comparison classification methods. Before feeding the HSIs into the GA classification network, each pixel should be converted to a GA multivector. The final results are obtained by averaging ten independent random sample runs.

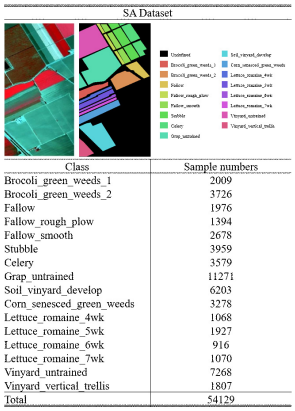
A. HSI Datasets

In this article, the performances of the proposed methods are evaluated on four publicly available HSI datasets with different settings. As shown in Fig. 3(a)–(d), the four datasets are Salinas Scene (SA), Indian Pines (IP), University of Pavia (PU), and Kennedy Space Center (KSC).

The first dataset is SA [Fig. 3(a)], which was captured by the airborne visible infrared imaging spectrometer (AVIRIS) instrument with 224-band over Salinas Valley, CA, USA. After discarding the low signal-to-noise ratio (SNR) bands, there are 204 bands with 3.7-m/pixel spatial resolution. The ground-truth image consists of 512 × 217 pixels and 16 classes can be identified.

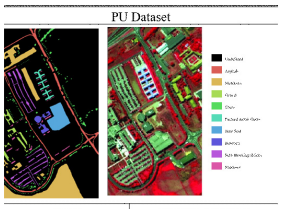
The second dataset is IP [Fig. 3(c)], which was collected by the AVIRIS in Northwestern Indiana, with 145 × 145 spatial dimension of 20 m/pixel and spectral bands of width 224 with wavelengths ranging from 0.4 to 2.5 μm. It has 200 bands available for distinguishing 16 different categories after removing bands covering the region of water absorption.

The third one is PU [Fig. 3(b)], which was acquired by the reflective optics system imaging spectrometer (ROSIS)-03 sensor over the campus at the University of Pavia, Pavia, Italy. The ROSIS sensor collects 115 bands and contains



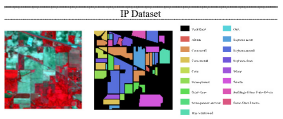
Class	Sample numbers
Brocoli_green_weeds_1	2009
Brocoli_green_weeds_2	3726
Fallow	1976
Fallow_rough_plow	1394
Fallow_smooth	2678
Stubble	3959
Celery	3579
Grap_untrained	11271
Soil_vinyard_develop	6203
Corn_senesced_green_weeds	3278
Lettuce_romaine_4wk	1068
Lettuce_romaine_5wk	1927
Lettuce_romaine_6wk	916
Lettuce_romaine_7wk	1070
Vinyard_untrained	7268
Vinyard_vertical_trellis	1807
Total	54129

(a)



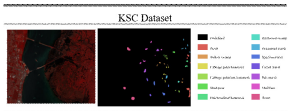
Class	Sample numbers
Asphalt	6631
Meadows	18649
Gravel	2099
Trees	3064
Painted metal sheets	1345
Bare Soil	5029
Bitumen	1330
Self-Blocking Bricks	3682
Shadows	947
Total	42776

(b)



Class	Sample numbers
Alfalfa	46
Corn-notill	1428
Corn-mintill	830
Corn	237
Grass-pasture	483
Grass-trees	730
Grass-pasture-mowed	28
Hay-windrowed	478
Oats	20
Soybean-notill	972
Soybean-mintill	2455
Soybean-clean	593
Wheat	205
Woods	1265
Buildings-Grass-Trees-Drives	386
Stone-Steel-Towers	93
Total	10249

(c)



Class	Sample numbers
Scrub	761
Willow swamp	243
Cabbage palm hammock	256
Cabbage palm/oak hammock	252
Slash pine	161
Oak/broadleaf hammock	229
Hardwood swamp	105
Graminoid marsh	431
Spartina marsh	520
Cattail marsh	404
Salt marsh	419
Mud flats	503
Water	927
Total	5211

(d)

Fig. 3. False-color composite image, ground-truth map, and a number of available samples of different datasets. (a) SA dataset. (b) PU dataset. (c) IP dataset. (d) KSC dataset.

610 × 340 pixels, covering the wavelength range from 430 to 860 nm (with 1.3 m/pixels). In the experiment, we utilize 103 spectral bands for the task of classification. Nine different land-cover classes are illustrated in the ground-truth map.

The last dataset KSC [Fig. 3(d)] was captured by the airborne AVIRIS sensor over KSC, Florida. It is composed of 512 × 614 pixels with a spatial resolution of 18 m/pixels, which has an altitude of approximately 20 km. After removing water absorption and low SNR bands, 176 bands can be used for identifying 13 classes. It is easy to recognize that the KSC dataset has the smallest size of samples.

B. Experimental Configurations

1) 1D-GACNN: The first part is designed to evaluate the performance of 1D-GACNN on the above four datasets in comparison to six different DL-based classification methods while only using spectral features. These six methods are k-nearest neighbor (KNN) [51], SVM with linear kernel (L-SVM), RBF-SVM, 1-D recurrent neural network (1D-RNN) [52], and multiple-layer perceptron (MLP) with one hidden layer composed of 2048 hidden neural units and 1D-RCNN [8]. 1D-GACNN has the same structure as 1D-RCNN, in detail, a GA convolutional layer, a GA maximum pooling layer, and two GA fully connected layers.

Furthermore, for 1D-GACNN, the learning rate and the optimizer for training are 0.01 and stochastic gradient

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TABLE I
CLASSIFICATION RESULTS WITH SPECTRAL FEATURES ON DIFFERENT DATASETS

Method	PU			SA			KSC			IP		
	OA(%)	AA(%)	K × 100	OA(%)	AA(%)	K × 100	OA(%)	AA(%)	K × 100	OA(%)	AA(%)	K × 100
KNN	79.11±0.84	79.84±0.61	72.88±0.96	85.50±0.44	92.36±0.11	83.83±0.49	86.09±0.50	79.32±0.65	84.52±0.56	72.45±0.40	66.89±1.75	68.35±0.46
L-SVM	87.41±0.48	86.50±0.37	83.39±0.58	91.36±0.43	96.02±0.17	90.32±0.47	88.97±0.48	82.37±1.44	87.71±0.54	80.99±0.82	81.99±1.44	78.50±0.92
RBF-SVM	89.52±0.72	88.57±0.53	86.16±0.90	90.12±0.53	94.93±0.16	88.94±0.59	91.48±1.41	87.37±1.98	90.52±1.57	82.47±0.36	83.14±1.01	80.16±0.41
1D-RNN	84.11±0.64	82.94±0.58	79.25±0.78	89.22±0.73	95.13±0.28	87.95±0.81	88.78±4.25	82.31±7.24	87.53±4.70	75.26±0.73	73.28±1.41	71.86±0.82
MLP	89.23±1.45	88.73±1.08	85.79±1.77	89.90±0.43	94.78±0.27	88.47±0.71	87.09±0.83	80.91±1.08	85.64±0.93	81.22±1.54	83.38±1.51	78.81±1.70
1D-RCNN	86.81±2.00	86.91±1.62	82.67±2.44	89.43±1.57	94.30±1.35	88.21±1.71	90.75±1.74	85.50±2.78	89.71±1.93	87.57±0.78	87.56±0.72	85.84±0.88
1D-GACNN	90.00±1.10	89.50±0.67	86.82±1.36	91.64±0.29	95.75±0.14	90.63±0.33	91.95±0.86	87.54±1.32	91.04±0.95	88.56±0.55	86.89±0.72	86.95±0.63

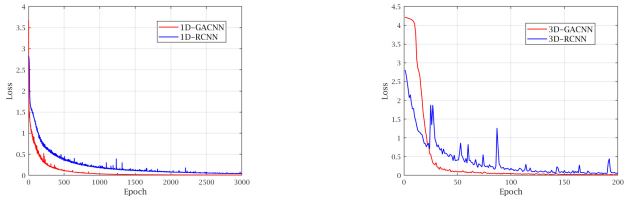
descent (SGD), respectively. For PU and SA, we randomly select 200 labeled pixels per class for training and all other pixels in the ground-truth map for testing, while for KSC and IP, 20% labeled pixels of the dataset are randomly selected for training and the remaining 80% for testing. The purpose of the division is to plenty show the performance of 1D-GACNN on datasets of different sizes.

2) 3D-GACNN: As for 3D-GACNN, we choose six deep neural networks that all use spectral–spatial techniques to extract high-level features for comparison. These methods can be divided into two categories, namely, traditional 3-D CNNs and state-of-the-art methods. The first category includes 3D-RCNN [13], miniGCN [53], SSCasRNN [54], and multiscale 3-D deep CNN (M3D-CNN) [15], while the second category consists of hybrid spectral net (HSN) [17] and S3EResBoF [20]. As stated in [13], 3D-GACNN has three GA convolutional layers, two GA maximum pooling layers, and one GA fully connected layer. Moreover, 50% of dropout is used to tackle the overfitting issue because the number of available training samples in experiments is limited.

For HSN and S3EResBoF, all datasets are processed by the PCA algorithm, and then, 15, 15, 30, and 80 spectral bands are used as input to PU, SA, IP, and KSC, respectively [17], [20]. Considering the redundant information of different bands and the structure of the network, for 3D-GACNN, 40, 40, 40, and 80 bands are selected as input for PU, SA, IP, and KSC after applying the PCA algorithm, respectively. The other four methods do not perform the operation of dimensionality reduction.

Based on the above configurations, the following three sets of dataset division are performed.

1) For PU and SA, we randomly select 100 labeled pixels per class for training and all other pixels in the ground-truth map for testing, while for KSC and IP, 10% labeled pixels of the dataset are randomly selected for training and the remaining 90% for testing (simply called the 100-0.1 division).
2) The number of training samples is increased to 200 per class for PU and SA, and 20% labeled pixels for IP and KSC. Similarly, the remaining samples in each dataset are used for testing in terms of OA, AA, and K (simply called the 200-0.2 division).
3) In order to verify that GA retains more correlation information among different channels of HSIs, we only use 50 random labeled pixels per class for PU and SA and 5% random labeled pixels of IP and KSC as training samples (simply called the 50-0.05 division).



(a) (b)

Fig. 4. (a) Loss of 1-D networks on IP. (b) Loss of 3-D networks on SA.

C. Experiment Results

We have tested two GA networks in our experiment, 1D-GACNN and 3D-GACNN. In Fig. 4, we can find that the performance of our GA-CNN is consistently better than that of real-valued CNN with the same structure. In detail, the training loss function of our GA-CNN converges more quickly than that of real-valued CNN and reaches a smaller loss finally. First, GA-CNN can suffer less from the risk of overfitting because the implicit regularizers imposed by the computation of multivectors. Second, the multivector convolution preserves more different information of HSIs, and this information is beneficial for classifying.

Next, we will discuss the specific classification results.

1) 1D-GACNN: Table I shows the classification results obtained by the comparison methods only using spectral information associated with each corresponding dataset. The classification accuracy in this article is given in the form of the mean and standard deviation. The best classification results are shown in bold. It is found that 1D-GACNN offers better performances in terms of OA and K on all four datasets. Under the same network structure, 1D-GACNN improves the OA of 1D-RCNN by 3.19%, 2.12%, 1.30%, and 0.99% on the PU, SA, KSC, and IP datasets, respectively. Although RBF-SVM offers similar accuracy with 1D-GACNN on the PU, SA, and KSC datasets, its performance on the IP dataset is severely degraded, while 1D-GACNN still maintains high classification accuracy, which indicates that the extensive applicability of 1D-GACNN is more extensive.

Figs. 5–8 show the ground-truth maps of the original image and different classification maps of the comparison classifiers, with the objective of evaluating the classification accuracy from a visual perspective. As can be seen from the blue region (bare soil) in Fig. 5, the 1D-GACNN method handles this category better. Similarly, Fig. 8 shows more accurate classification to Woods and Oats by using 1D-GACNN. In general, the proposed 1D-GACNN exhibits the best classification

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TABLE II
CLASSIFICATION RESULTS WITH SPECTRAL–SPATIAL FEATURES ON THE KSC DATASET USING 10% TRAINING SAMPLES

Method	3D-RCNN	SSCasRNN	miniGCN	M3D-CNN	HSN	S3EResBoF	3D-GACNN
OA(%)	94.92±1.33	95.86±0.24	91.66±0.61	89.71±2.76	96.52±0.83	92.91±5.12	97.14±1.38
AA(%)	93.84±1.39	95.38±0.38	86.55±0.99	84.70±3.25	94.40±1.32	92.25±4.05	96.13±1.71
K × 100	94.34±1.48	95.74±0.26	90.72±0.68	88.55±3.08	96.12±0.93	92.11±5.70	96.82±1.53
Scrub	96.91±2.22	95.79±0.74	96.58±0.45	97.76±0.87	96.68±1.51	94.94±5.83	98.48±1.10
Willow swamp	86.73±5.08	95.59±1.06	90.69±2.38	88.70±3.78	93.89±4.21	88.71±11.01	95.68±7.50
Cabbage palm hammock	91.93±2.72	93.12±2.05	76.83±4.02	77.26±4.61	90.21±5.34	80.73±16.35	93.99±5.12
Cabbage palm/oak hammock	89.23±4.12	89.1.6±2.97	52.94±3.17	56.07±8.11	79.39±6.93	76.42±17.33	86.82±6.07
Slash pine	96.93±1.77	93.34±2.17	65.90±4.18	58.72±7.95	85.06±9.00	84.61±11.56	91.65±6.06
Oak/broadleaf hammock	91.69±4.52	95.18±1.50	76.86±4.61	79.70±4.97	92.61±3.45	94.69±11.93	97.33±2.26
Hardwood swamp	96.19±3.91	96.09±1.48	86.21±5.51	86.18±4.73	93.76±6.69	98.27±3.03	98.27±2.08
Graminoid marsh	91.77±3.72	92.34±1.70	92.27±1.36	82.16±7.00	98.38±1.00	87.17±16.03	96.47±7.42
Spartina marsh	94.27±2.79	96.53±2.43	96.74±0.93	91.51±3.66	99.73±0.31	99.49±0.67	99.24±0.77
Cattail marsh	93.49±2.94	96.88±1.52	96.41±1.06	92.28±7.76	99.73±0.46	95.50±3.28	98.01±1.37
Salt marsh	95.47±2.73	96.91±0.29	97.72±1.77	98.85±0.59	99.94±0.10	99.95±0.16	97.81±1.59
Mud flats	95.43±2.18	96.95±1.74	96.16±1.17	92.18±5.41	97.91±1.38	98.78±1.67	96.00±1.94
Water	99.84±0.23	96.97±0.18	99.84±0.14	99.71±0.48	99.86±0.22	100±0.00	99.93±0.10

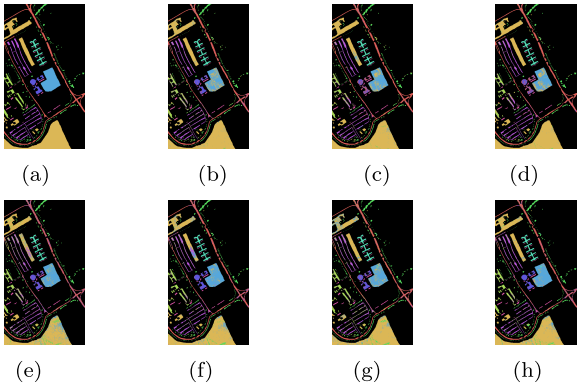


Fig. 5. Classification results of the comparison methods on the PU dataset. (a) Ground-truth map. (b) KNN. (c) L-SVM. (d) RBF-SVM. (e) 1D-RNN. (f) MLP. (g) 1D-RCNN. (h) 1D-GACNN.

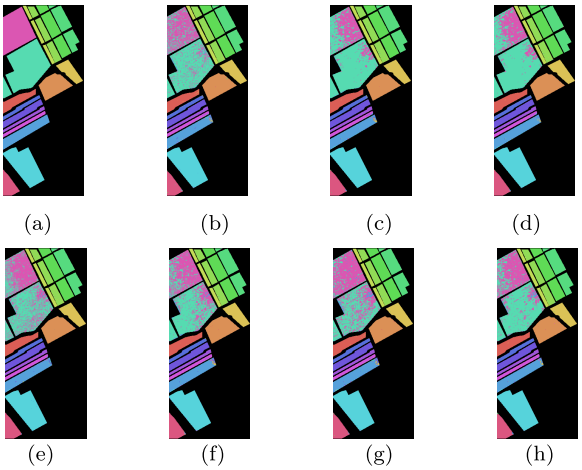


Fig. 6. Classification results of the comparison methods on the SA dataset. (a) Ground-truth map. (b) KNN. (c) L-SVM. (d) RBF-SVM. (e) 1D-RNN. (f) MLP. (g) 1D-RCNN. (h) 1D-GACNN.

results compared to the other comparison classical classification methods, which demonstrates the effectiveness of the extension to the GA domain.

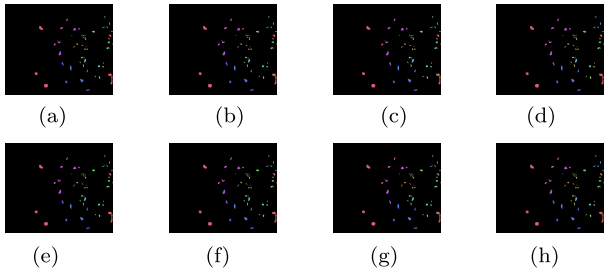


Fig. 7. Classification results of the comparison methods on the KSC dataset. (a) Ground-truth map. (b) KNN. (c) L-SVM. (d) RBF-SVM. (e) 1D-RNN. (f) MLP. (g) 1D-RCNN. (h) 1D-GACNN.

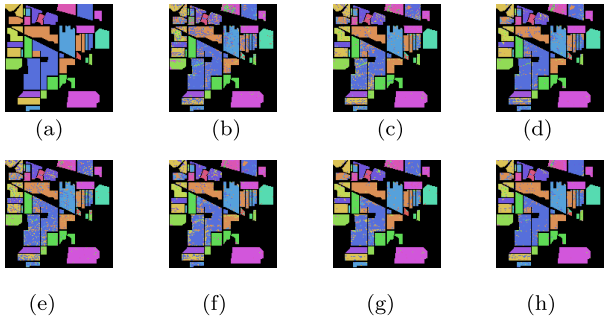


Fig. 8. Classification results of the comparison methods on the IP dataset. (a) Ground-truth map. (b) KNN. (c) L-SVM. (d) RBF-SVM. (e) 1D-RNN. (f) MLP. (g) 1D-RCNN. (h) 1D-GACNN.

2) 3D-GACNN: Next, we consider the performance of 3D-GACNN, which extracts both spectral and spatial features (i.e., pixel and its neighborhoods as inputs). The size of the window is set to 27 × 27 for 3D-GACNN in all experiments. The first experiment is based on the 100-0.1 division and the classification can be found in Tables II–V. As can be observed from the tables, 3D-GACNN achieves better results on the IP, KSC, and SA datasets. Although the OA, AA, and K of 3D-GACNN are 0.33%, 0.59%, and 0.43% lower than HSN on the PU dataset, respectively, its standard deviation for ten random experiments is smaller, which means that

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TABLE III
CLASSIFICATION RESULTS WITH SPECTRAL–SPATIAL FEATURES ON THE IP DATASET USING 10% TRAINING SAMPLES

Method	3D-RCNN	SSCasRNN	miniGCN	M3D-CNN	HSN	S3EResBoF	3D-GACNN
OA(%)	93.04±2.38	94.22±0.44	93.99±0.40	89.70±1.67	95.75±2.87	97.02±0.79	97.53±0.41
AA(%)	91.46±2.26	93.54±0.93	93.86±0.94	90.29±1.91	92.56 ±4.91	95.08±2.59	95.90±1.18
K × 100	92.09±2.67	94.03±0.50	93.15±0.45	88.31±1.91	95.17 ± 3.24	96.61±0.91	97.18±0.31
Alfalfa	95.21±5.16	94.65±1.41	86.77±9.53	86.21±7.11	87.91 ±12.91	94.79±6.42	92.71±4.23
Corn-notill	88.77±4.33	93.69±0.71	92.48±0.93	88.85±2.26	93.64 ±7.17	96.41±1.78	96.32±1.19
Corn-mintill	90.15±6.46	94.01±1.11	90.55±0.73	84.18±2.04	94.97 ±2.56	96.52±4.82	97.17±1.03
Corn	92.59±3.65	93.67±1.35	90.10±2.50	87.10±4.50	89.64 ±11.85	95.88±3.91	97.57±1.83
Grass-pasture	93.10±2.18	93.77±0.97	97.37±0.71	96.37±1.29	95.12 ±2.51	95.18±3.35	96.96±1.43
Grass-trees	96.09±1.10	94.54±0.56	98.72±0.41	98.30±0.84	96.97 ±2.60	98.77±0.66	97.23±0.85
Grass-pasture-mowed	89.97±7.99	90.67±7.28	89.97±5.95	86.81±8.25	87.62 ±13.09	76.98±39.05	96.75±5.01
Hay-windrowed	99.30±1.03	95.46±0.14	99.03±0.89	98.60±0.72	99.13 ±1.40	99.82±0.37	99.72±0.42
Oats	64.44±18.99	89.41±6.78	96.86±3.95	84.47±9.07	71.81 ±19.96	98.41±3.31	82.31±10.48
Soybean-notill	90.76±1.43	93.09±1.25	90.81±1.45	83.03±4.08	95.26 ±2.92	96.37±2.61	97.09±0.80
Soybean-mintill	93.39±2.67	94.56±0.39	93.91±0.52	86.41±2.59	97.30 ±1.29	97.54±1.03	98.24±0.54
Soybean-clean	89.32±3.35	93.44±1.11	91.58±1.54	85.71±3.71	93.51 ±2.94	95.99±2.18	95.56±1.11
Wheat	95.17±2.84	94.45±0.76	98.89±0.55	97.86±2.41	96.70 ±2.49	96.80±2.55	98.28±1.61
Woods	98.07±1.00	95.25±0.23	97.19±0.64	96.57±0.27	98.31 ±1.85	98.44±2.16	99.26±0.46
Buildings-Grass-Trees-Drives	96.07±1.67	94.72±0.54	88.76±1.60	87.91±2.29	94.76 ±2.90	98.55±1.30	97.77±1.40
Stone-Steel-Towers	90.98±3.65	91.28±3.23	98.73±0.65	96.29±2.24	88.30 ±7.32	84.87±9.85	91.42±4.26

TABLE IV
CLASSIFICATION RESULTS WITH SPECTRAL–SPATIAL FEATURES ON THE SA DATASET USING 100 TRAINING SAMPLES IN EACH CLASS

Method	3D-RCNN	SSCasRNN	miniGCN	M3D-CNN	HSN	S3EResBoF	3D-GACNN
OA(%)	92.26±2.33	93.95±1.43	92.13±0.50	91.77±0.98	98.90 ±1.60	98.37±0.30	99.21±0.25
AA(%)	95.35±1.35	95.23±0.72	96.58±0.50	96.13±0.44	99.29 ± 1.04	99.16±0.31	99.54±0.13
K × 100	91.38±2.58	93.59±1.59	91.22±0.55	90.84±1.08	98.77 ±1.78	98.18±0.70	99.11±0.28
Brocoli_green_weeds_1	97.07±2.40	95.87±0.90	99.84±0.14	99.95±0.05	99.99±0.02	99.93±0.19	99.92±0.14
Brocoli_green_weeds_2	97.64±1.88	96.29±0.40	99.90±0.10	99.85±0.17	99.93±0.13	100±0.00	99.93±0.13
Fallow	98.08±1.85	96.52±0.39	98.95±0.64	98.93±0.55	99.95±0.10	99.97±0.10	99.98±0.03
Fallow_rough_plow	98.55±0.79	96.27±0.41	99.49±0.50	99.37±0.36	98.82±1.83	97.75±2.22	98.87±0.53
Fallow_smooth	99.10±0.79	96.67±0.24	99.12±0.56	99.34±0.30	99.73±0.15	99.75±0.44	99.25±0.63
Stubble	99.50±0.47	96.58±0.42	99.95±0.08	99.95±0.06	99.85±0.31	99.96±0.04	99.95±0.06
Celery	98.49±1.00	96.33±0.39	99.71±0.17	99.74±0.19	99.85±0.35	99.99±0.01	99.89±0.23
Grap_untrained	83.98±5.70	90.77±3.35	81.62±1.12	80.21±3.80	97.44±4.24	98.09±2.01	98.36±0.56
Soil_vinyard_develop	98.06±1.36	96.41±0.63	99.73±0.17	99.40±0.28	99.97±0.04	100±0.00	99.93±0.12
Corn_senesced_green_weeds	96.97±1.78	95.74±0.81	95.73±2.29	93.10±2.18	98.50±1.71	99.57±0.63	99.20±0.92
Lettuce_romaine_4wk	96.59±2.02	95.57±1.85	97.72±1.42	97.48±1.49	97.93±5.15	99.80±0.28	99.93±0.08
Lettuce_romaine_5wk	93.44±3.91	95.24±1.55	99.71±0.38	98.91±0.93	99.62±0.45	99.92±0.14	99.91±0.10
Lettuce_romaine_6wk	94.15±3.65	95.94±0.77	99.77±0.25	98.93±0.79	99.95±0.11	99.62±0.95	99.94±0.10
Lettuce_romaine_7wk	96.96±2.62	95.39±1.29	98.96±1.38	97.62±0.88	99.70±0.31	99.67±0.54	99.76±0.35
Vinyard_untrained	78.51±7.57	88.65±3.73	75.81±1.74	76.05±2.40	97.48±3.39	92.63±4.59	97.86±0.96
Vinyard_vertical_trellis	98.45±0.98	95.42±0.46	99.48±0.32	99.20±0.33	99.94±0.10	99.97±0.09	99.97±0.08

TABLE V
CLASSIFICATION RESULTS WITH SPECTRAL–SPATIAL FEATURES ON THE PU DATASET USING 100 TRAINING SAMPLES IN EACH CLASS

Method	3D-RCNN	SSCasRNN	miniGCN	M3D-CNN	HSN	S3EResBoF	3D-GACNN
OA(%)	92.40±1.34	93.68±0.79	93.77±0.51	93.11±1.52	98.69±1.40	97.68±1.43	98.36±0.30
AA(%)	91.73±1.57	93.21±0.68	92.19±0.51	93.20±1.64	98.36±1.70	96.63±1.80	97.77±0.36
K × 100	90.02±1.73	92.56±1.04	91.75±0.66	90.93±1.95	98.24±1.89	96.92±1.88	97.81±0.40
Asphalt	90.82±3.06	93.58±0.78	91.68±1.14	95.54±1.34	99.17±0.71	98.71±0.99	97.16±1.17
Meadows	94.99±1.10	94.87±0.89	97.39±0.62	94.53±1.34	99.31±0.72	99.86±0.12	99.49±0.28
Gravel	84.33±4.77	89.81±2.23	82.44±1.99	90.67±3.29	97.22±3.17	92.03±11.39	97.52±1.62
Trees	95.62±1.57	93.46±1.31	96.83±0.87	93.89±3.35	96.66±3.08	89.14±10.35	96.97±1.06
Painted metal sheets	99.14±1.08	96.92±0.07	99.98±0.04	99.80±0.19	99.78±0.26	99.10±0.62	99.61±0.44
Bare Soil	86.36±3.52	90.63±2.49	91.27±1.54	84.61±2.81	98.69±1.48	98.73±1.26	99.21±0.71
Bitumen	87.69±3.78	91.51±1.36	83.60±2.76	90.68±3.55	99.25±0.78	99.25±0.54	98.25±1.64
Self-Blocking Bricks	92.83±1.73	94.09±1.03	87.40±1.18	93.05±1.68	96.09±6.85	96.35±1.51	95.24±0.99
Shadows	93.78±4.83	94.05±2.25	99.14±0.51	96.07±3.65	99.11±0.45	96.47±2.05	96.46±1.37

3D-GACNN can be more stable in dealing with random samples. The reason is that the HSN model basically combines the complementary information of spatiospectra and spectra in the form of 3-D and 2-D convolutions, respectively. The HSN achieves the highest accuracy in Table V at the expense of a larger number of model parameters. The 3D-GACNN

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TABLE VI
COMPARISON OF MODEL PARAMETERS OF SEVERAL METHODS

	Dataset	Method						
		3D-RCNN	SSCasRNN	miniGCN	M3D-CNN	HSN	S3EResBoF	3D-GACNN
Total parameters	PU	1068266	75808	105047	89306	4844922	217673	16840
	IP	1123281	134008	146496	284897	5122305	246800	22468
	SA	1125457	135932	148168	289249	4845825	218128	22468
	KSC	1102254	114013	136077	208606	7812990	349005	132056
Parameter size (MB)	PU	4.08	0.29	0.40	0.34	18.48	0.83	0.064
	IP	4.28	0.51	0.56	1.09	19.54	0.94	0.086
	SA	4.29	0.52	0.57	1.1	18.49	0.83	0.086
	KSC	4.2	0.43	0.51	0.8	29.8	1.33	0.504

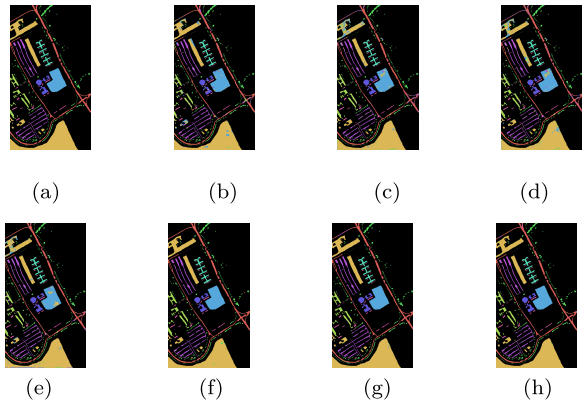


Fig. 9. Classification results of the comparison methods on the PU dataset. (a) Ground-truth map. (b) 3D-RCNN. (c) SSCasRNN. (d) miniGCN. (e) M3D-CNN. (f) HSN. (g) S3EResBoF. (h) 3D-GACNN.

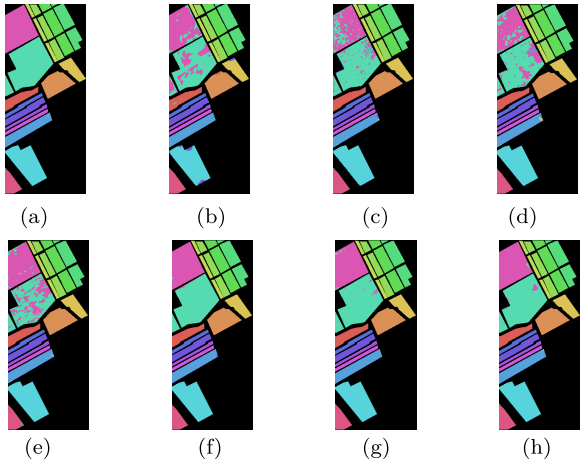


Fig. 10. Classification results of the comparison methods on the SA dataset. (a) Ground-truth map. (b) 3D-RCNN. (c) SSCasRNN. (d) miniGCN. (e) M3D-CNN. (f) HSN. (g) S3EResBoF. (h) 3D-GACNN.

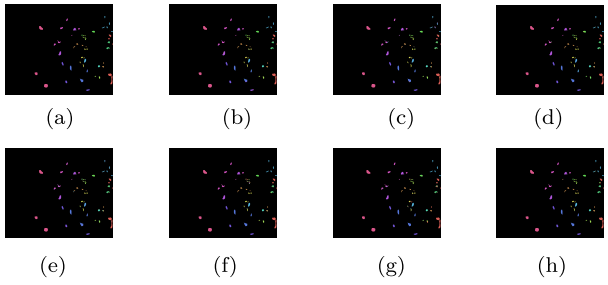


Fig. 11. Classification results of the comparison methods on the KSC dataset. (a) Ground-truth map. (b) 3D-RCNN. (c) SSCasRNN. (d) miniGCN. (e) M3D-CNN. (f) HSN. (g) S3EResBoF. (h) 3D-GACNN.

achieves slightly lower classification accuracy compared with the HSN, but the model parameters of the HSN are 200 times more than those of the 3D-GACNN, which can be found in Table VI. It is worth mentioning that S3EResBoF, which performs well on the PU, SA, and IP datasets, achieves a great reduction in classification accuracy on the smallest dataset, KSC (see Table II). Similarly, on the IP dataset (see Table III), which includes many categories but few samples, HSN also fails to attain the high accuracy. Because multivector preserves the correlation between various spectral channels, the classification ability of 3D-GACNN is robust to a variety of HSIs in practical applications.

The visualization of land-cover classification maps is compared using the training model with the mean OA, as shown in Figs. 9–12 for PU, SA, KSC, and IP datasets, respectively. It can be seen from Figs. 9 and 10 that the classification ability of 3D-GACNN is comparable to HSN and S3EResBoF, and the three methods are much stronger than the other four methods [Figs. 9(b)–(e) and 10(b)–(e)]. For small datasets with multiple classifications (see Fig. 12), the proposed 3D-GACNN achieves the highest sensitivity and specificity compared with the other competing methods, e.g., buildings-Grass-Trees-Drives, Soybean-mintill, and Corn-notill.

In order to evaluate the performance of the proposed network comprehensively, the third experiment uses the 50-0.05 division just for HSN, S3EResBoF, and 3D-GACNN due to the aforementioned poor experimental results achieved by the other four methods. Such a division rule means that there are only 512 training samples in the IP dataset with 16 classes and 216 training samples in the KSC dataset with 13 classes. The OA, AA, and K with these standard deviations (std) of the three methods for four datasets are presented in Figs. 13–15, respectively. For the IP dataset, the classification accuracy of HSN is severely reduced because the number of training samples cannot meet the requirements of the network parameters. At the same time, compared with 3D-GACNN, S3EResBoF also has a poor performance on the KSC dataset,

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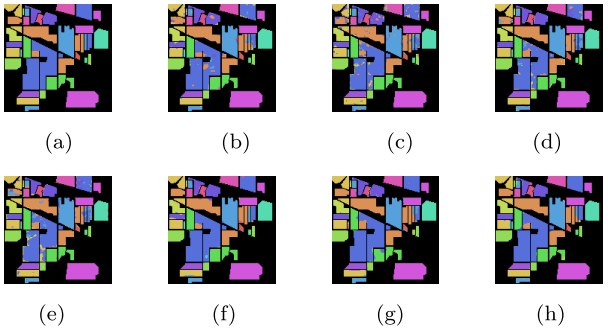


Fig. 12. Classification results of the comparison methods on the IP dataset. (a) Ground-truth map. (b) 3D-RCNN. (c) SSCasRNN. (d) miniGCN. (e) M3D-CNN. (f) HSN. (g) S3EResBoF. (h) 3D-GACNN.

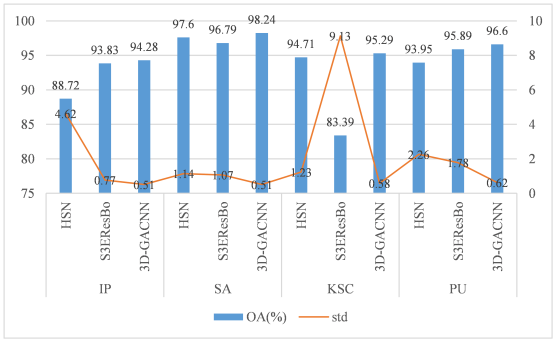


Fig. 13. OA for 3D-GACNN, S3EResBoF, and HSN on four datasets.

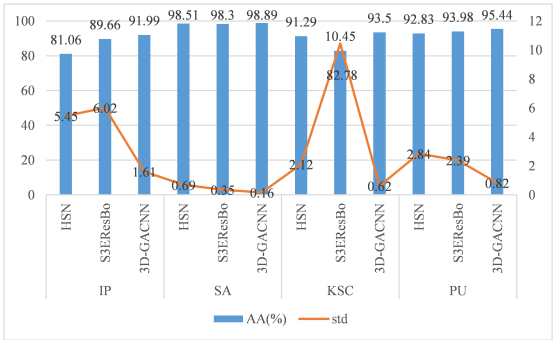


Fig. 14. AA for 3D-GACNN, S3EResBoF, and HSN on four datasets.

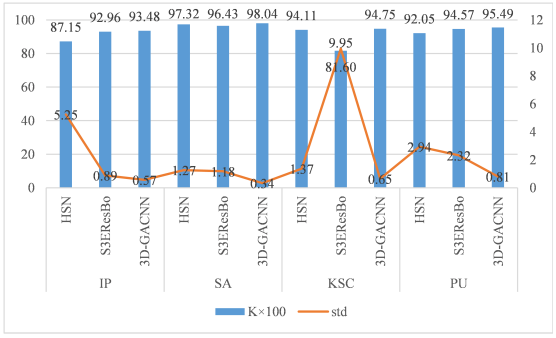


Fig. 15. K for 3D-GACNN, S3EResBoF, and HSN on four datasets.

which results in a 11.9%, 10.72%, and 13.15% reduction in OA, AA, and K, respectively. For the SA and PU datasets, with slightly more training samples, HSN and S3EResBoF attain the accuracy over 90%, which is still lower than that

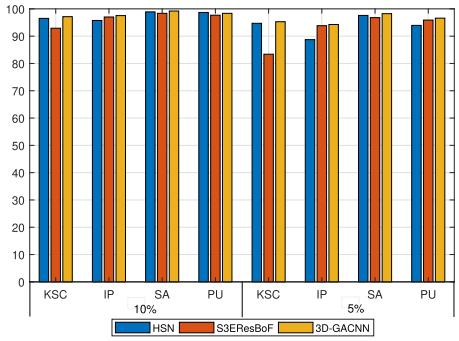


Fig. 16. OA for HSN, S3EResBoF, and 3D-GACNN with different ratios of training samples.

of 3D-GACNN. In addition, we can observe that the standard deviations of OA, AA, and K for 3D-GACNN on the four datasets are the smallest, indicating that 3D-GACNN has a very obvious advantage in classification stability.

Fig. 16 visually reflects the impact of the training sample ratio on classification accuracy. Specifically, 5% and 10% labeled pixels per class for PU and SA, and 5% and 10% labeled pixels of IP and KSC as training samples. As can be seen from Fig. 16, when the training samples are sufficient, the classification performance of 3D-GACNN and S3EResBoF is better than that of HSN. When the ratio of training samples decreases, although the classification results of all algorithms deteriorate, the classification performance of 3D-GACNN is still better than that of the other two comparison algorithms.

It can be concluded from the classification results that 3D-GACNN can achieve good classification accuracy when there are a few training samples, and the reason for this is its fewer model parameters. For a fair comparison, the trainable parameters of 3D-GACNN are the real numbers for training stored in the memory. For example, a 2-D multivector is stored as four real numbers, and a 3-D multivector is stored as eight real numbers. As shown in Table VI, 3D-GACNN requires the least number of trainable parameters except on the KSC dataset, that is, the model only occupies a relatively small storage space. The minimum parameters are shown in bold. In particular, compared to HSN and S3EResBoF, which perform well in the above experiments, the training network of 3D-GACNN is more lightweight. For example, for the PU dataset, the memory capacity of HSN is 288 times that of 3D-GACNN, while S3EResBoF is 13 times.

In addition, the training time for experiment one can be found in Table VII, where the minimum training time is indicated in bold. Since most of the comparison methods relevant to the 1D-GACNN are not neural networks based, the parameters are not comparable. As for the running time, there is little difference. 3D-RCNN has the longest training time on each dataset, especially SA and IP, because it uses all bands for classification and does not use enough convolutional and pooling layers of different sizes to reduce the spectral dimension in the classification network [12], [14], [15], resulting in a lot of time to process much redundant information. In general, apart from 3D-RCNN, S3EResBoF requires more training time because it uses the spectral and spatial residual learning blocks

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TABLE VII
TRAINING TIME BASED ON 100-0.1 DIVISION FOR DIFFERENT METHODS

	Dataset	Method						
		3D-RCNN	SSCasRNN	miniGCN	M3D-CNN	HSN	S3EResBoF	3D-GACNN
Training time (min)	PU	46.68	1.43	1.82	1.88	5.90	8.45	19.48
	IP	95.67	2.14	2.97	5.41	10.12	18.05	14.04
	SA	77.43	2.95	4.14	5.57	10.02	17.07	12.50
	KSC	34.56	1.18	1.21	2.51	8.54	16.40	12.26

TABLE VIII
CLASSIFICATION RESULTS OF SEVERAL METHODS

Dataset		Method							
		BHCNN-STE	BHCNN-CFL	DGCN	DGCN-D	DGCN-C	DGCN-M	DGCN-DC	3D-GACNN
IP	OA(%)	96.10±0.70	97.40±0.90	73.50±5.00	75.58±2.58	77.06±2.09	78.92±1.83	77.46±2.89	94.28±0.51
	AA(%)	92.20±1.30	94.10±1.60	70.70±5.43	74.78±1.95	75.83±1.88	79.34±1.80	77.79±2.33	91.99±1.61
	K × 100	95.60±1.00	97.00±1.40	67.81±6.39	71.72±2.80	74.65±2.11	76.68±2.02	77.63±3.14	93.48±0.57
PU	OA(%)	99.70±0.20	99.82±0.20	75.60±6.00	77.21±4.49	78.28±3.36	78.40±4.24	79.87±5.74	96.60±0.62
	AA(%)	99.50±0.30	99.63±0.20	77.05±5.98	79.80±4.47	78.46±3.35	78.43±4.73	79.84±5.81	95.44±0.82
	K × 100	99.60±0.20	99.72±0.20	72.75±6.77	74.26±5.07	75.46±3.77	75.61±4.78	77.09±6.52	95.49±0.81

that consume more computational resources. It is worth noting that the training time of 3D-GACNN with the least trainable parameters in average is not the shortest. This is because the noncommutative multiplication rule in GA leads to more time for carrying out convolution operations for multivectors.

Essentially, our GA-CNN adds an implicit regularizer during the processing, which ensures the representations of the HSI under the guidance of multivector operations. Such a strategy takes more complicated relationships across different channels into account and suppresses the degrees of freedom of model's parameters during training. For this reason, using GA-CNN, we can achieve better learning results with fewer parameters compared with real-valued CNNs. In addition, an HSI is represented, circulated, and processed in the form of a multivector matrix in our model rather than several separate channels, which preserves more color information than real-valued CNN. Experimental results show that our GA-CNN model provides benefits for HSI classification, which outperforms its competitors.

3) Further Analysis: In order to study the classification methods of HSIs more comprehensively, this part uses the 50-0.05 division for BHCNN-STE, BHCNN-CFL [55], DGCN, DGCN-D, DGCN-C, DGCN-M, DGCN-DC [56], and 3D-GACNN. The OA, AA, and K with these standard deviations (std) of these methods for two datasets are presented in Table VIII. As can be seen from Table VIII, the classification results of the BHCNN-style methods and 3D-GACNN are much better than those achieved by the DGCN-style methods. In addition, the classification results of the BHCNN-style methods are slightly better than those of the 3D-GACNN. This is because the BHCNN-style methods connect an auxiliary classifier after the band selection layer, which further optimizes the classification performance of the BHCNN-style methods.

In the BHCNN-STE and BHCNN-CFL methods, they expand the receptive fields by constructing 3-D dilated convolutions and extract spatial–spectral features. The 3D-GACNN uses GA to innovate in feature extraction and represents different band images as GA multivectors to capture the inherent structures and preserve the correlation of those channels, so as to improve the classification performance of the model. The main reason why the classification performance of the BHCNN-STE and BHCNN-CFL methods is better than that of the 3D-GACNN is that the former adds the band selection process and the auxiliary classification layer to the training process, which can effectively carry out network training, optimize the final classifier and auxiliary classifier, and thus obtain a better classification performance. At present, the 3D-GACNN focuses on innovation and improvement in feature extraction. With regard to the BHCNN method, we can try to add band selection and the auxiliary classifier to the 3D-GACNN in our subsequent research to further improve the performance of the 3D-GACNN.

V. CONCLUSION

In this article, new CNNs based on GA (GA-CNN) was proposed as an extension of CNN from the real number domain to the GA domain for dealing with multichannel HSIs. Essentially, each spectrum in an HSI is mapped to a component of the geometric algebraic multivector, and then, the inherent structures and the correlation of those channels are captured and preserved based on the geometric algebraic multivector. The GA-CNN can preserve the inherent structural information among the spectral bands to the greatest extent when processing the HSI, which improves the processing efficiency of the HSI in contrast to the insufficiency of the real-valued CNN in a high-dimensional space. The proposed GA-CNN was described in detail, including convolution, pooling, and fully connected layers. Also, we derived the BP training algorithm, in which the weights are represented and updated in the GA space. To evaluate the flexibility of our model, classification experiments were carried out on four widely

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used datasets. The experimental results demonstrated that the proposed GA-CNN shows a significant improvement in the classification performance compared to the existing traditional and state-of-the-art methods, especially when there are a few available training samples. At the same time, 3D-GACNN is able to greatly reduce the number of parameters of the network while maintaining high classification accuracy. Nevertheless, GA-CNN also has a small drawback. The noncommutative multiplication of GA may increase the computational complexity. With the development of GA parallel computing, it is expected that the proposed GA-CNN will become faster. More importantly, this article opens a new research direction by showing that GA-CNN has the potential to become a powerful tool in multidimensional signal processing.

Appendix
Proof of the GA Layers

In GA-BP, the weights are updated as follows:

$$\begin{aligned}\triangle\, w_{\mathrm{kj}}^{(L)} &= \sum_{A\in\mathbf{b}} \frac{\partial E}{\partial\left[w_{kj}^{(L)}\right]_A} e_A \\ &= \sum_{C\in\mathbf{b}} \frac{\partial E}{\partial\left[s_j^{(L)}\right]_C} \frac{\partial\left[s_j^{(L)}\right]_C}{\partial\left[w_{kj}^{(L)}\right]_A}. \end{aligned} \tag{22}$$

According to the chain rule, we obtain $\delta_j{}^{(L)}$ as follows:

$$\begin{aligned}\delta_j^{(L)} &= \sum_{C\in\mathbf{b}} \frac{\partial E}{\partial\left[s_j^{(L)}\right]_C} e_C \\ &= \sum_{C\in\mathbf{b}} \frac{\partial E}{\partial\left[y_j\right]_C} \frac{\partial\left[y_j^{(L)}\right]_C}{\partial\left[\mathrm{s}_j^{(L)}\right]_C} e_C \\ &= \left(\left[d_j\right]_C - \left[y_j^{(L)}\right]_C\right)\left(\mathrm{g}^{(L)}\right)'\left(\left[\mathrm{s}_j^{(L)}\right]_C\right) e_C. \end{aligned} \tag{23}$$

Then, the weights in the fully connected layer L are updated as follows:

$$\begin{aligned}\triangle\, w_{\mathrm{kj}}^{(L)} &= \sum_{A\in\mathbf{b}} \frac{\partial E}{\partial\left[w_{kj}^{(L)}\right]_A} e_A \\ &= \sum_{A\in\mathbf{b}} \left(\sum_{C\in\mathbf{b}} \left[\delta_{\mathrm{j}}^{(L)}\right] \frac{\partial\left[s_j^{(L)}\right]_C}{\partial\left[w_{kj}^{(L)}\right]_A}\right) e_A \\ &= \left(d_j - y_j^{(L)}\right) \odot \left(g^{(L)}\right)'\left(s_j^{(L)}\right) \otimes_{p,q} \left(x_k^{(L-1)}\right)^* \\ \triangle\, \theta_j^{(L)} &= \delta_j^{(L)}. \end{aligned} \tag{24}$$

For two multivectors $W = \sum_{A\in\mathbf{b}} \mathrm{w}_A e_A = \mathrm{w}_0 e_0 + \mathrm{w}_1 e_1 + \cdots + \mathrm{w}_n e_n$ and $Z = \sum_{A\in\mathbf{b}} \mathrm{z}_A e_A = \mathrm{z}_0 e_0 + \mathrm{z}_1 e_1 + \cdots + \mathrm{z}_n e_n$ with the same dimension, we define $W \odot Z = \mathrm{w}_0 z_0 e_0 + \mathrm{w}_1 z_1 e_1 + \cdots + \mathrm{w}_n z_n e_n$.

The weights in the fully connected layers l ($l < L$) are updated as follows:

$$\delta_j^{(l)} = \left(\sum_j \left(w_{kj}^{(l+1)}\right)^* \cdot \delta_j^{(l+1)}\right) \odot \left(g^{(l)}\right)'\left(\left[s_j^{(l)}\right]_C\right) e_C$$

$$\begin{aligned}\triangle\, w_{kj}^{(l)} &= \delta_j^{(l)} \cdot \left(x_k^{(L-1)}\right)^* \\ &= \left(\sum_j \left(w_{kj}^{(l+1)}\right) \cdot \delta_j^{(l+1)}\right) \\ &\quad \odot \left(g^{(l)}\right)'\left(\left[s_j^{(l)}\right]\right) \cdot \left(x_k^{(L-1)}\right)^*. \end{aligned} \tag{25}$$

Similar to the bias, we have

$$\triangle\, \theta_j^{(l)} = \delta_j^{(l)}. \tag{26}$$

The error of the pooling layers can be derived as follows:

$$\begin{aligned}\delta_j^{(l)} &= \left(g^{(l)}\right)'\left(s^{(l)}\right) \\ &\quad \times \mathrm{conv2}_G\left(\delta_j^{(l+1)}, \mathrm{rot180}\left(\mathrm{w}_j^{(l+1)}\right), \text{'full'}\right) \\ \triangle\, \theta_j^{(l)} &= \sum_j \delta_j^{(l)} \end{aligned} \tag{27}$$

where $\mathrm{conv2}_G(\cdot)$ represents the convolution of multivector matrix, $\mathrm{rot180}(\cdot)$ represents an operator that aims to make the convolution function perform cross correlation with the rotated kernel, and "full" represents an operator that pads the missing inputs with zeros automatically.

The error of the convolutional layers can be derived as follows:

$$\delta^{(l)} = \mathrm{rot180}\left(\mathrm{w}_j^{(l)}\right)^* \cdot elta^{(l+1)} \odot \left(g^{(l)}\right)'\left(\left[\mathrm{s}_j^{(l)}\right]_C\right) e_C \tag{28}$$

where $\mathrm{rot180}(\cdot)$ represents the rotation of the jth convolution kernel $\mathrm{w}_j^{(l)}$ in the convolutional layers.

Similarly, the weights in the convolutional layers are updated as follows:

$$\begin{aligned}\triangle\, w^{(l+1)} &= \delta^{(l)} \cdot \mathrm{rot180}\left(x_k^{(l-1)}\right) \\ &= \left(\sum_j \left(\mathrm{w}_j^{(l)}\right)^* \cdot \delta^{(l-1)}\right) \\ &\quad \odot \left(g^{(l)}\right)'\left(s_j^{(l)}\right) \cdot \left(x_k^{(l-1)}\right)^*. \end{aligned} \tag{29}$$

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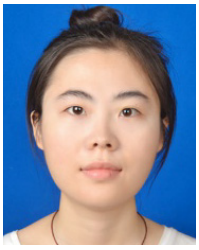
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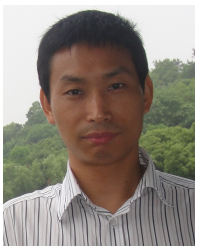
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